Fashion MNIST Image Classification Using Principal Component Analysis (PCA) and Multi-Layer Perceptron (MLP) Neural Network

Dataset

- Fashion MNIST Dataset
- 60,000 Training Images
- 10,000 Test Images
- 28 × 28 grayscale images
- 784 original features per image

Preprocessing

- Min-Max Normalization using MinMaxScaler
- Pixel values scaled between 0 and 1

PCA Configuration

Parameter	Value
Original Features	784
PCA Components	50
Reduced Features	50
Dimensionality Reduction	93.62%

The dimensionality was reduced from 784 features to only 50 principal components, significantly decreasing computational complexity.

Neural Network Architecture

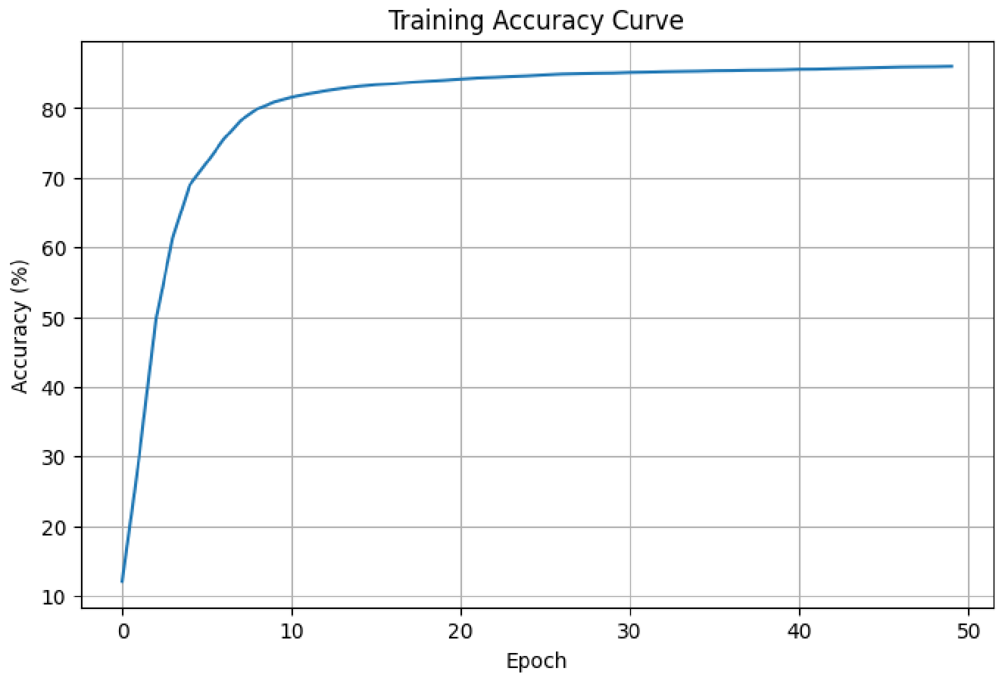
Your network is a custom-built MLP implemented from scratch using backpropagation.

Layer	Neurons	Activation
Input Layer	50	PCA Features
Hidden Layer 1	15	Sigmoid
Hidden Layer 2	16	Sigmoid
Output Layer	10	Sigmoid

Training Parameters

Parameter	Value
Learning Rate	0.01
Epochs	50
Optimizer	Gradient Descent (Custom Backpropagation)
Loss Function	Sum of Squared Error
Random Seed	1

Training Performance Analysis



The network showed steady convergence during training.

Initial Training Accuracy

- Epoch 0: 12.04%

Final Training Accuracy

- Approximately 84–85%

Observation

The loss decreased consistently across epochs while training accuracy increased steadily, indicating effective learning and proper convergence of the model.

The absence of sharp fluctuations suggests stable gradient updates and successful optimization.

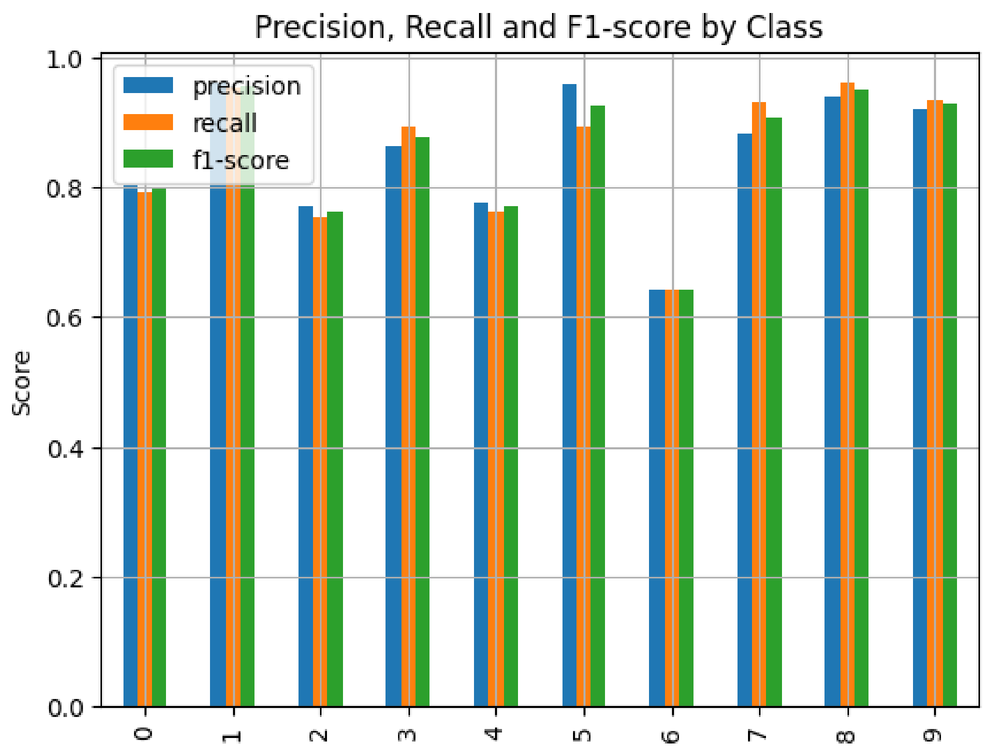
Test Performance

Overall Test Accuracy

85.00%

This demonstrates that the model generalizes reasonably well to unseen Fashion MNIST images despite substantial dimensionality reduction.

Classification Report Analysis



Overall Metrics

Metric	Score
Accuracy	0.85
Macro Precision	0.85

Metric	Score
Macro Recall	0.85
Macro F1-score	0.85

Best Performing Classes

Trouser (Class 1)

- Precision: 0.96
- Recall: 0.95
- F1 Score: 0.96

The model easily distinguishes trousers because of their unique shape and pixel distribution.

Bag (Class 8)

- Precision: 0.94
- Recall: 0.96
- F1 Score: 0.95

Bags possess distinct visual characteristics that are effectively preserved in PCA components.

Ankle Boot (Class 9)

- Precision: 0.92
- Recall: 0.94
- F1 Score: 0.93

Footwear categories retain strong discriminative features even after dimensionality reduction.

Weakest Performing Class

Shirt (Class 6)

- Precision: 0.64
- Recall: 0.64
- F1 Score: 0.64

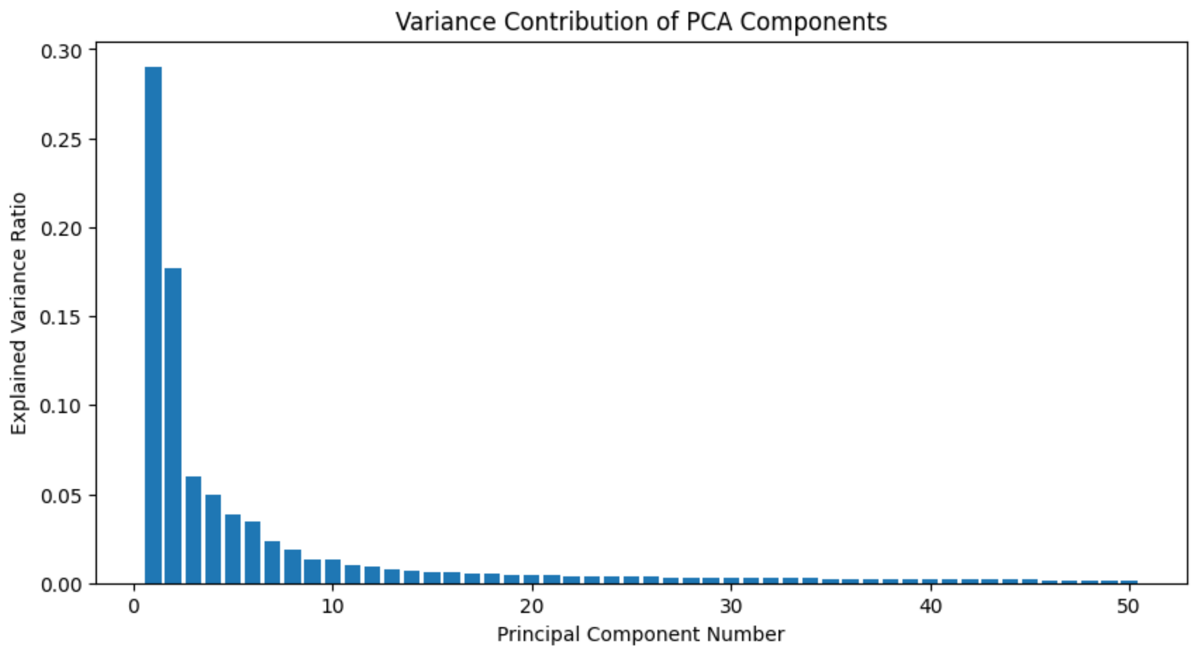
Shirts share visual similarities with:

- T-Shirts
- Pullovers
- Coats

This overlap causes confusion and lowers classification performance.

Interpretation of PCA Results

Explained Variance Plot



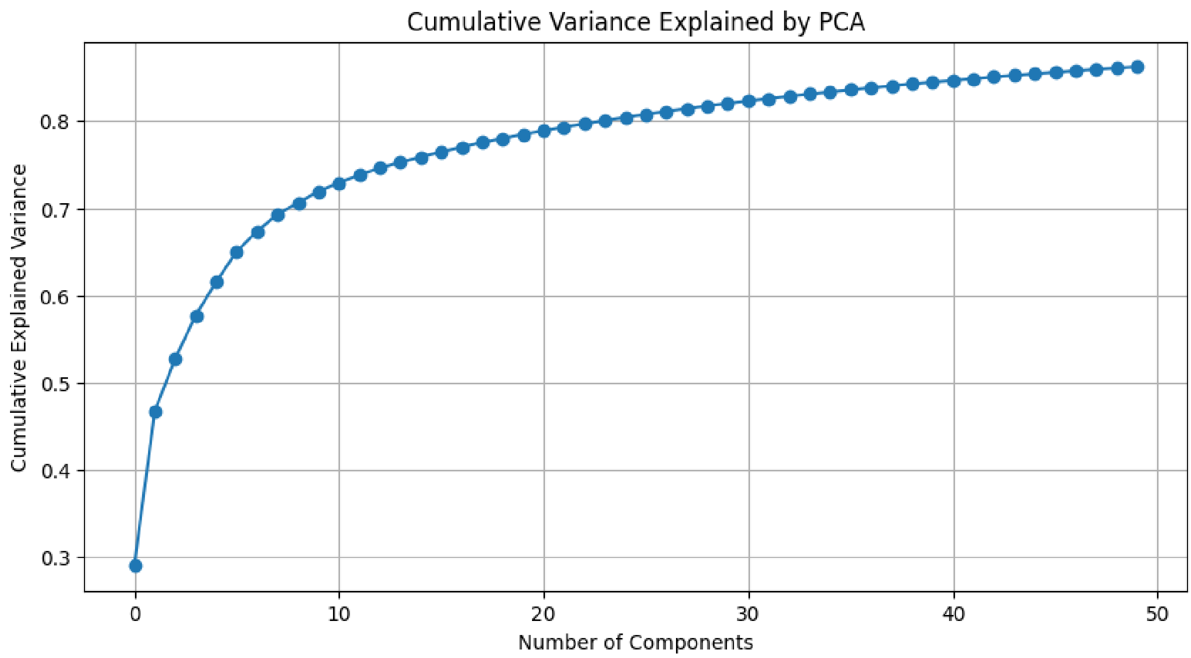
The variance contribution graph demonstrates that early principal components capture the majority of information contained within the original dataset.

The first few components contain dominant structural patterns such as:

- Object outlines
- Shape information
- Edge distributions

Subsequent components contribute progressively less information.

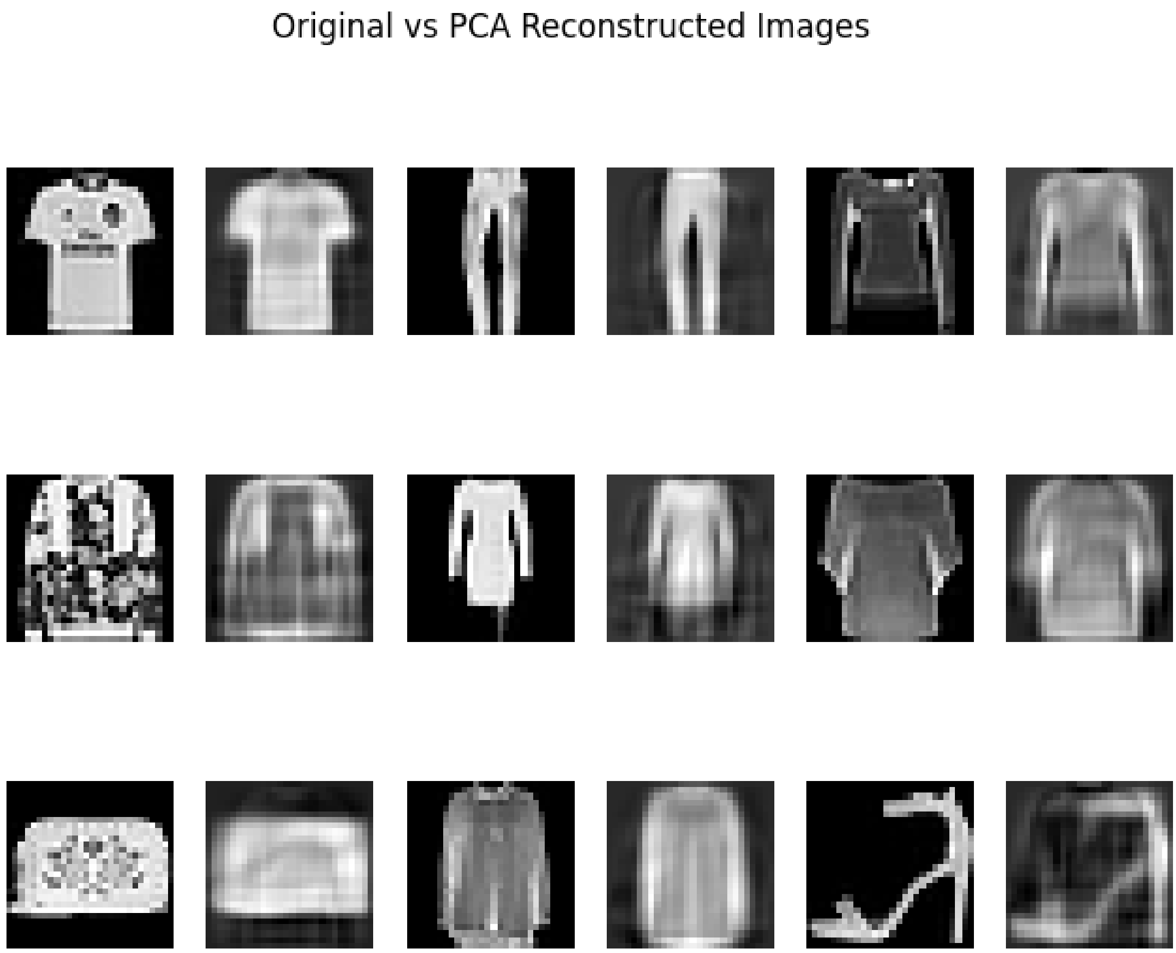
Cumulative Variance Plot



The cumulative variance curve rises rapidly and gradually stabilizes, indicating that most meaningful information is retained within the first 50 components.

This validates the choice of PCA for dimensionality reduction.

PCA Reconstruction Analysis

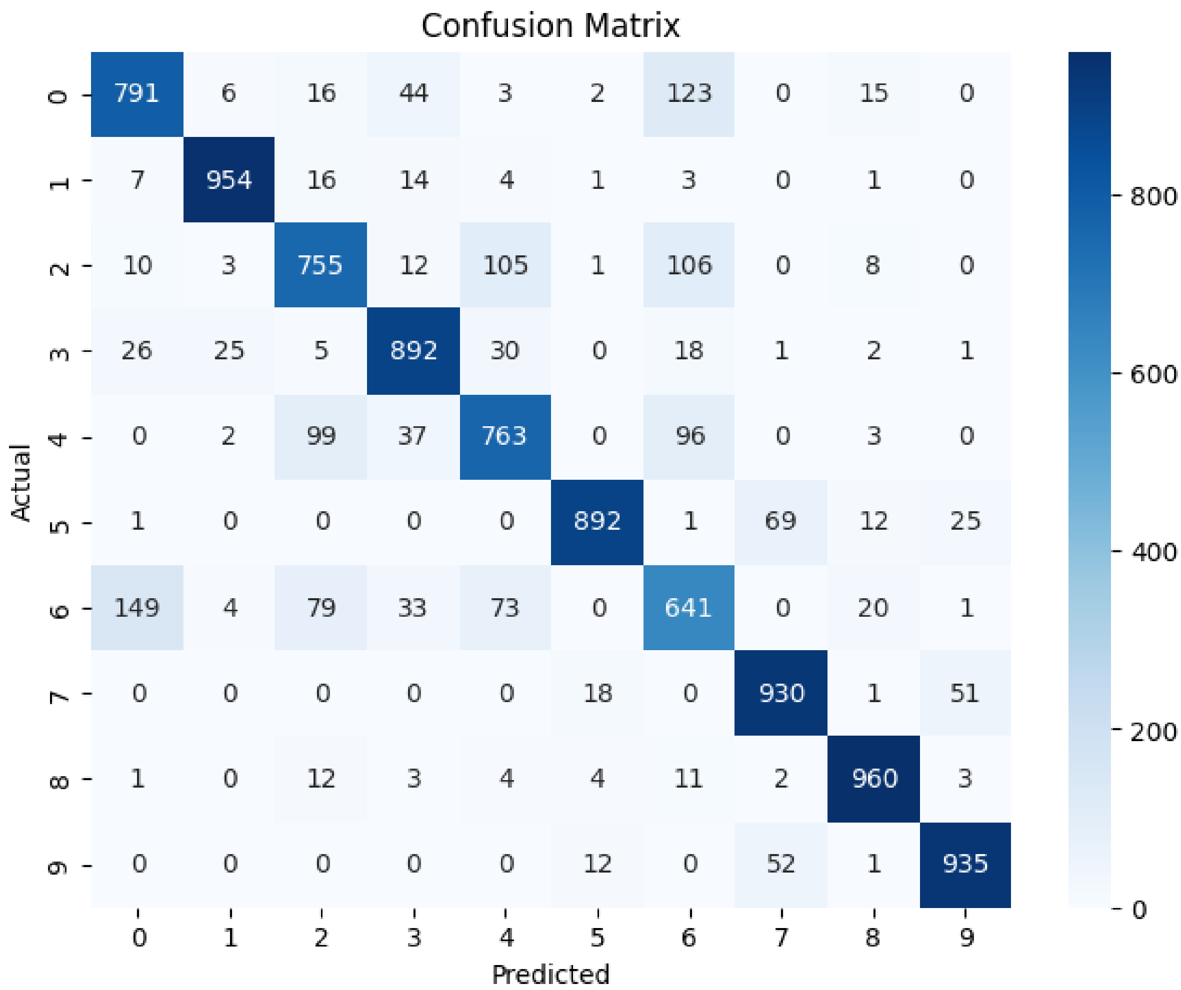


The reconstructed images generated through inverse PCA transformation show that:

- Major object structures are preserved.
- Fine details are slightly blurred.
- Essential class-specific features remain visible.

This confirms that PCA successfully compresses the dataset while retaining meaningful visual information.

Confusion Matrix Interpretation



The confusion matrix reveals:

High Classification Accuracy

Classes such as:

- Trouser
- Bag
- Sneaker
- Ankle Boot

show strong diagonal dominance, indicating correct classification.

Common Misclassifications

The model occasionally confuses:

Actual	**Predicted**
Shirt	T-Shirt

Actual	Predicted
Coat	Pullover
Pullover	Shirt
Dress	Coat

These classes exhibit similar visual textures and silhouettes.

Storytelling and Business Insight

Overall Conclusion

This study proves that significant dimensionality reduction can be achieved without severely compromising classification performance.

The combination of PCA and MLP offers:

- Reduced computational cost
- Faster training
- Lower memory consumption
- Good classification accuracy
- Improved model efficiency

Therefore, PCA serves as an effective preprocessing technique for neural-network-based image classification tasks where computational resources are limited.

From a business perspective, this approach can support:

- Fashion recommendation systems
- Automated product tagging
- Visual search engines
- Inventory classification
- Retail analytics

where computational efficiency is critical.